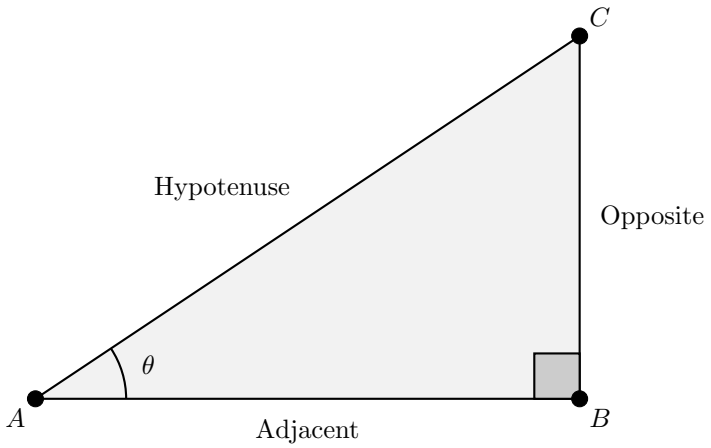
# 1 Foundational Definitions




Let $\triangle ABC$ be a right triangle with $\angle B = 90°$, and let $\theta = \angle BAC$ denote the angle at vertex $A$. Thus, the hypotenuse is the side $AC$, and the segment $AB$ is adjacent and $BC$ opposite to $\theta$. The trigonometric functions are defined as the following:

**Definition 1** (Sine)**.**

$$\sin\theta = \frac{BC}{AC} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

**Definition 2** (Cosecant)**.**

$$\csc\theta = \frac{AC}{BC} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

**Definition 3** (Cosine)**.**

$$\cos\theta = \frac{AB}{AC} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

**Definition 4** (Secant)**.**

$$\sec\theta = \frac{AC}{AB} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

**Definition 5** (Tangent)**.**

$$\tan\theta = \frac{BC}{AB} = \frac{\text{Opposite}}{\text{Adjacent}}$$

**Definition 6** (Cotangent)**.**

$$\cot\theta = \frac{AB}{BC} = \frac{\text{Adjacent}}{\text{Opposite}}$$

# 2 Immediate Algebraic Properties

## 2.1 Reciprocal Identities

**Identity 1.**

$$\sin\theta = \frac{1}{\csc\theta}$$

**Identity 2.**

$$\cos\theta = \frac{1}{\sec\theta}$$

**Identity 3.**

$$\tan\theta = \frac{1}{\cot\theta}$$

**Identity 4.**

$$\csc\theta = \frac{1}{\sin\theta}$$

**Identity 5.**

$$\sec\theta = \frac{1}{\cos\theta}$$

**Identity 6.**

$$\cot\theta = \frac{1}{\tan\theta}$$

### 2.2 Quotient Identities

## 3 Cofunction Identities

**Definition 7** (Complementary Angles)**.** Two angles are *complementary* if and only if the sum of their measures is equal to the measure of a right angle ($90^\circ$ or $\frac{\pi}{2}$ radians).

Formally, let $\alpha$ and $\beta$ be the measures of two angles. They are complementary if and only if: $\alpha + \beta = 90^\circ$. Either angle is referred to as the complement of the other.

**Theorem 1** (Congruence of Complements)**.** *If two angles are complementary to the same angle, then they are congruent to each other. Similarly, if they are complementary to two congruent angles, they are congruent.*

*Proof.* Let angle $\alpha$ and angle $\beta$ be complementary to a common angle $\gamma$.

By the definition of complementary angles, $\alpha + \gamma = 90^\circ$ and $\beta + \gamma = 90^\circ$.

By the transitive property of equality, we have

$$\alpha + \gamma = \beta + \gamma.$$

Subtracting $\gamma$ from both sides yeilds

$$\alpha = \beta.$$

Thus, $\alpha \cong \beta$. □

**Theorem 2** (Complementarity of Acute Angles in a Right Triangle)**.** *The two acute angles of any right-angled triangle are complementary.*

*Proof.* Let $\triangle ABC$ be a right triangle with $\angle B = 90^\circ$. The sum of the interior angles of any triangle is $180^\circ$. Thus,

$$\angle A + \angle B + \angle C = 180^\circ.$$

Substituting $\angle B = 90^\circ$, we have

$$m(\angle A) + m(\angle B) + 90^\circ = 180^\circ$$

Subtracting $90^\circ$ from both sides gives

$$\angle A + \angle C = 90^\circ.$$

By definition, since the sum of $m(\angle A)$ and $m(\angle C)$ equals $90^\circ$, $\angle A$ and $\angle C$ are complementary. □

**Identity 7** (Cofunction Identities)**.**

$$\sin(90^\circ - \theta) = \cos\theta, \quad \cos(90^\circ - \theta) = \sin\theta$$

*Proof.* Let $\triangle ABC$ be a right triangle with $\angle B = 90^\circ$. □

# 4 Pythagorean Identities

**Theorem 3** (Pythagorean Theorem)**.** *For right triangle $\triangle ABC$, the lengths of the sides satisfy:*

$$AB^2 + BC^2 = AC^2$$

**Identity 8** (Pythagorean Identity)**.**

$$\sin^2\theta + \cos^2\theta = 1$$

*Proof.* Using the side-length definitions of sine and cosine for $\triangle ABC$:

$$\sin\theta = \frac{BC}{AC} \quad \text{and} \quad \cos\theta = \frac{AB}{AC}$$

Squaring both terms gives:

$$\sin^2\theta = \frac{BC^2}{AC^2} \quad \text{and} \quad \cos^2\theta = \frac{AB^2}{AC^2}$$

Adding them together:

$$\sin^2\theta + \cos^2\theta = \frac{BC^2}{AC^2} + \frac{AB^2}{AC^2} = \frac{AB^2 + BC^2}{AC^2}$$

By the Pythagorean Theorem, $AB^2 + BC^2 = AC^2$, which yields:

$$\sin^2\theta + \cos^2\theta = \frac{AC^2}{AC^2} = 1$$

□

**Identity 9** (Pythagorean Identity)**.**

$$\sec^2\theta - \tan^2\theta = 1$$

*Proof.* From the Pythagorean Identity, divide both sides by $\cos^2\theta$, yielding

$$\begin{aligned} \sin^2\theta + \cos^2\theta = 1 &\implies \frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \\ &\implies \frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta} \end{aligned}$$

By the definitions of tangent and secant, we have

$$\begin{aligned} \frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta} &\implies \tan^2\theta + 1 = \sec^2\theta \\ &\implies \sec^2\theta - \tan^2\theta = 1 \end{aligned}$$

□

**Identity 10** (Pythagorean Identity)**.**

$$\csc^2\theta - \cot^2\theta = 1$$

*Proof.* From the Pythagorean Identity, divide both sides by $\sin^2\theta$, yielding

$$\begin{aligned}\sin^2\theta+\cos^2\theta=1 &\implies \frac{\sin^2\theta}{\sin^2\theta}+\frac{\cos^2\theta}{\sin^2\theta}=\frac{1}{\sin^2\theta}\\ &\implies 1+\frac{\cos^2\theta}{\sin^2\theta}=\frac{1}{\sin^2\theta}\end{aligned}$$

By the definition of cotangent and cosecant, we have

$$\begin{aligned}1+\frac{\cos^2\theta}{\sin^2\theta}=\frac{1}{\sin^2\theta} &\implies 1+\cot^2\theta=\csc^2\theta\\ &\implies \csc^2\theta-\cot^2\theta=1\end{aligned}$$

□

# 5 Angle Sum and Difference

**Identity 11.**

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

**Identity 12.**

$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$$

**Identity 13.**

$$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$$

# 6 Double-Angle Identities

# 7 Half-Angle and Power Reducing

# 8 Even/Odd Identities